Cessna 182T Skylane (C182T)

Lycoming IO-540-AB1A5, 230 hp · 4 seats · Typical 182T (2001+) POH procedures; fuel-injected engine, constant-speed propeller, cowl flaps. G1000 (2004+) wording noted where it differs.

V-speeds & limits

Vr	**50–60 KIAS**
Vx	**65 KIAS**
Vy	**80 KIAS**
Best glide	**75 KIAS (3,100 lb)**
Vs / Vso	**49 / 41 KIAS**
Vfe	**140 KIAS (10°) / 100 KIAS (10°–20°)**
Va	**110 KIAS (3,100 lb) / 98 KIAS (2,500 lb) / 89 KIAS (2,000 lb)**
Vno	**140 KIAS**
Vne	**175 KIAS**
Approach	**70–80 KIAS flaps up / 60–70 KIAS flaps full / 60 KIAS short field**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Fuel quantity	**CHECK visually (87 gal usable)**
Oil quantity	**CHECK (9 qt max, 5 qt min)**
Passenger briefing	**COMPLETE (belts, doors, exits, sterile cockpit)**
Seats, belts, harnesses	**ADJUST & LOCK**
Brakes	**TEST & SET**
Control lock	**REMOVE**
Circuit breakers	**CHECK IN**
Electrical equipment / autopilot	**OFF**
Avionics switch (BUS 1 & BUS 2 on G1000)	**OFF**
Fuel selector	**BOTH**
Fuel shutoff valve	**ON (push full in)**
Cowl flaps	**OPEN**
Standby battery	**TEST (20 s, green steady) then ARM (G1000)**
Master switch (ALT & BAT)	**ON**
Annunciators / PFD	**CHECK (LOW VOLTS on is normal before start)**

Engine Start

Throttle	**OPEN 1/4 inch**
Mixture	**IDLE CUTOFF**
Propeller	**HIGH RPM (full forward)**
Beacon	**ON**
Propeller area	**CLEAR**
Fuel pump	**ON**
Mixture	**RICH until fuel flow stable (3–5 s), then IDLE CUTOFF**
Fuel pump	**OFF**
Ignition switch	**START (release when engine fires)**
Mixture	**ADVANCE smoothly to RICH when engine fires**
Throttle	**ADJUST 1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60 s cold)**
Ammeter / LOW VOLTS annunciator	**CHECK charging / light out**
Avionics switch (BUS 1 & BUS 2 on G1000)	**ON**

Before Taxi

Altimeter / barometric setting	**SET**
Radios / transponder / GPS	**SET**
Flaps	**CHECK operation, then UP**
Parking brake	**RELEASE**
Brakes	**CHECK during taxi**
Flight instruments	**CHECK during turns (HSI / DG, turn coordinator, compass)**

Before Takeoff

Parking brake	**SET**
Seats, belts, harnesses	**CHECK SECURE**
Cabin doors	**CLOSED & LATCHED**
Flight controls	**FREE & CORRECT**
Flight instruments / PFD	**CHECK & SET (no red X)**
Fuel quantity	**CHECK**
Fuel selector	**BOTH**
Mixture	**RICH (below 3,000 ft; lean for field elevation above)**
Elevator / rudder trim	**TAKEOFF**
Cowl flaps	**OPEN**
Throttle	**1,800 RPM**
Magnetos	**CHECK (max drop 150 RPM, max 50 RPM differential)**
Propeller	**CYCLE from high to low RPM 2–3 times (RPM drop ~300–500, pressure recovery)**
Engine instruments / ammeter	**CHECK (oil press 50–90 psi green)**
Vacuum gauge (analogue)	**CHECK (4.5–5.5 in Hg)**
Annunciators	**CHECK (none lit)**
Throttle	**1,000 RPM**
Autopilot	**OFF**
Radios / transponder	**SET / ALT**
Strobe lights	**ON**
Wing flaps	**0°–20° (20° for short field)**
Takeoff briefing	**COMPLETE**

After Takeoff / Climb

Airspeed	**80 KIAS (Vy) / 65 KIAS (Vx, obstacles); cruise climb 85–95 KIAS**
Wing flaps	**UP (above 70 KIAS)**
Power	**23 in Hg / 2,400 RPM (full throttle, 2,400 RPM max continuous)**
Mixture	**RICH (lean above 3,000 ft per fuel flow placard)**
Cowl flaps	**OPEN**
Engine instruments	**CHECK (CHT / oil temp)**
Fuel pump	**OFF**

Cruise

Power	**15–23 in Hg, 2,000–2,400 RPM (80% max continuous)**
Elevator / rudder trim	**ADJUST**
Mixture	**LEAN as required (per fuel flow / EGT)**
Cowl flaps	**CLOSED / AS REQUIRED for CHT**
Engine instruments	**CHECK**
Fuel quantity / endurance	**CHECK**

Descent

Altimeter / barometric setting	**SET**
Power	**AS DESIRED (avoid rapid cooling – reduce 1–2 in Hg per minute)**
Mixture	**ADJUST for smooth operation**
Cowl flaps	**CLOSED**
Approach briefing	**COMPLETE**

Before Landing

Seats, belts, harnesses	**SECURE**
Fuel selector	**BOTH**
Mixture	**RICH**
Propeller	**HIGH RPM (full forward)**
Cowl flaps	**CLOSED (OPEN if go-around likely in hot conditions)**
Landing / taxi lights	**ON**
Autopilot	**OFF**
Airspeed	**70–80 KIAS (flaps up)**
Wing flaps	**AS DESIRED (10° below 140 KIAS, 10°–20° below 100 KIAS)**
Airspeed	**60–70 KIAS (flaps full)**

After Landing

Wing flaps	**UP**
Cowl flaps	**OPEN**
Strobe / landing lights	**OFF / AS REQUIRED**

Shutdown

Parking brake	**SET**
Throttle	**IDLE (1,000 RPM then idle)**
Electrical equipment / autopilot / pitot heat	**OFF**
Avionics switch (BUS 1 & BUS 2 on G1000)	**OFF**
Mixture	**IDLE CUTOFF (pull full out)**
Ignition switch	**OFF (key removed)**
Master switch (ALT & BAT)	**OFF**
Standby battery	**OFF (G1000)**
Control lock	**INSTALL**
Fuel selector	**LEFT or RIGHT (prevents crossfeed on sloped ramp)**

Abnormal procedures

LOW VOLTS / Alternator Failure

LOW VOLTS annunciator or ammeter discharge in flight

Avionics switch (BUS 1 & BUS 2)	**OFF**
Master switch (ALT only)	**OFF, then ON**
LOW VOLTS annunciator / ammeter	**CHECK**
Avionics switch (BUS 1 & BUS 2)	**ON**

▶ Charging restored?

Yes:

Ammeter / volts	**MONITOR**

No:

Master switch (ALT only)	**OFF**
Non-essential electrical equipment	**OFF (lights, autopilot, BUS 2 avionics)**
Standby battery	**CHECK ARMED – approx 30 min essential bus (G1000)**
Deferred: Alternator failed – battery / standby battery only, limited endurance. Land as soon as practical, lower flaps early while power remains.	**NOTED**

Low Oil Pressure

OIL PRESSURE annunciator or pressure below 50 psi green arc

Oil pressure / oil temperature	**CHECK**

▶ Oil temperature also rising?

Yes:

Power	**REDUCE to minimum required**
Propeller	**EXPECT governor loss – RPM may go to high pitch stop**
Engine failure	**EXPECT – plan a precautionary landing**
Deferred: Low oil pressure with high temperature – land as soon as possible, expect engine failure.	**NOTED**

No – temp normal:

Flight	**CONTINUE to nearest suitable airport, monitor**
Deferred: Low oil pressure indication – land as soon as practical, maintenance check.	**NOTED**

High Oil / Cylinder Head Temperature

Oil temperature or CHT in red arc

Cowl flaps	**OPEN**
Mixture	**RICH**
Power	**REDUCE**
Airspeed	**INCREASE (lower nose for cooling)**
Oil pressure	**MONITOR**

▶ Temperature returning to green arc?

Yes:

Flight	**CONTINUE, monitor**

No:

Deferred: High engine temperature – land as soon as practical, minimum power, cowl flaps open.	**NOTED**

Rough Running / Fuel Flow Fluctuation

Rough engine, fuel flow fluctuating (vapour) or power loss

Fuel pump ..... **ON**

Mixture ..... **ADJUST for smooth operation**

Fuel selector ..... **BOTH**

▶ Running smooth?

Yes:

Fuel pump ..... **OFF when fuel flow stable (ON again if fluctuation returns)**

No:

Magnetos ..... **CHECK L / R / BOTH, select smoothest**

Alternate induction air ..... **OPENS AUTOMATICALLY (no action)**

Deferred: Rough running engine – land as soon as practical, be ready for engine failure. ..... **NOTED**

Propeller Overspeed

RPM above 2,400 and not responding to propeller control

Throttle ..... **RETARD**

Oil pressure ..... **CHECK**

Propeller control ..... **DECREASE RPM (full aft; set if governor responds)**

Airspeed ..... **REDUCE**

Throttle ..... **AS REQUIRED to keep RPM below 2,400**

Deferred: Propeller governor failed – RPM controlled with throttle and airspeed only, land as soon as practical. ..... **NOTED**

Flap Failure

Flap circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Approach ..... **FLAPS UP – 70–80 KIAS, flatter approach, longer runway**

Deferred: Flaps inoperative – flaps-up landing at 70–80 KIAS, increased landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **REDUCE (approx 80 KIAS)**

Aircraft ..... **CONTROL – door has little effect on handling**

Window on open-door side ..... **OPEN (equalise pressure), then door CLOSE & LATCH**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Deferred: Door unlatched – land as soon as practical, normal landing, keep speed low. ..... **NOTED**

Vacuum / AHRS / PFD Failure

Vacuum gauge low or gyro flags (analogue); red X on PFD or display blank (G1000)

▶ Panel type?

G1000:

Display backup (red button) ..... **PRESS – reversionary mode on remaining display**

Standby airspeed / attitude / altimeter ..... **USE and cross-check**

Standby battery ..... **CHECK ARMED**

Deferred: G1000 display / AHRS failed – reversionary mode or standby instruments; avoid IMC. ..... **NOTED**

Analogue:

Vacuum gauge ..... **CHECK (4.5–5.5 in Hg)**

Attitude / heading indicators ..... **DISREGARD – cross-check**

Turn coordinator, compass, altimeter, airspeed ..... **USE for partial-panel flight**

Deferred: Vacuum failed – attitude and heading indicators unreliable; VMC only. ..... **NOTED**

IMC ..... **AVOID / EXIT**

Inadvertent Icing Encounter

Pitot heat ..... **ON**

Exit icing conditions ..... **TURN BACK / CHANGE ALTITUDE**

Cabin heat / windshield defrost ..... **FULL ON**

Power ..... **INCREASE / propeller RPM INCREASE to minimise propeller ice**

Mixture ..... **ADJUST (alternate induction air opens automatically)**

Airspeed on approach ..... **INCREASE (flaps up or 10° max with residual ice)**

Deferred: Residual airframe ice – flaps-up / 10° landing, approach 10–15 KIAS faster, expect higher stall speed. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

On the runway

Throttle (memory)	IDLE
Brakes (memory)	APPLY
Wing flaps (memory)	RETRACT
Mixture	IDLE CUTOFF
Ignition switch	OFF
Master switch (ALT & BAT)	OFF

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	75 KIAS flaps up / 70 KIAS flaps 10°–full
Mixture (memory)	IDLE CUTOFF
Fuel shutoff valve (memory)	OFF (pull full out)
Ignition switch (memory)	OFF
Wing flaps (memory)	AS REQUIRED (full recommended)
Master switch (ALT & BAT) (memory)	OFF
Cabin door	UNLATCH

Land straight ahead with only small changes of direction. Do not attempt to turn back to the runway.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	75 KIAS (best glide)
Propeller (memory)	LOW RPM (full aft – reduces drag)
Fuel shutoff valve (memory)	ON (push full in)
Fuel selector (memory)	BOTH
Fuel pump (memory)	ON
Mixture (memory)	RICH (if restart not immediate: IDLE CUTOFF, then advance slowly)
Ignition switch (memory)	BOTH (START if propeller stopped)

▶ Engine restarted?

Yes:

Propeller	HIGH RPM, then as required
Power	AS REQUIRED
Fuel pump	OFF (ON if power fluctuates)
Engine instruments	MONITOR
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system or ignition.	NOTED

No – forced landing:

Forced landing without engine power:

Field / wind	SELECT & PLAN (into wind if possible)
Mayday / 7700	TRANSMIT (time permitting)
Seats, belts, harnesses	SECURE
Airspeed	75 KIAS flaps up / 70 KIAS flaps down
Mixture	IDLE CUTOFF
Fuel shutoff valve	OFF (pull full out)
Ignition switch	OFF
Wing flaps	AS REQUIRED (full recommended)
Master switch (ALT & BAT)	OFF (when landing assured)
Doors	UNLATCH prior to touchdown
Touchdown	SLIGHTLY TAIL LOW
Brakes	APPLY HEAVILY

Engine Fire During Start

Cranking (memory) ..... CONTINUE (draw flames into engine)

▶ **Engine started?**

Yes:

Power ..... 1,800 RPM for a few minutes

Engine ..... SHUT DOWN & INSPECT for damage

No:

Throttle ..... FULL OPEN

Mixture ..... IDLE CUTOFF

Cranking ..... CONTINUE

Fuel pump ..... OFF

Fuel shutoff valve ..... OFF

Ignition switch ..... OFF

Master switch (ALT & BAT) ..... OFF

Fire extinguisher ..... OBTAIN & USE

Aircraft ..... EVACUATE, inspect for damage

Engine Fire in Flight

LAND IMMEDIATELY

Mixture (memory) ..... IDLE CUTOFF

Fuel shutoff valve (memory) ..... OFF (pull full out)

Fuel pump (memory) ..... OFF

Master switch (ALT & BAT) (memory) ..... OFF

Cabin heat / air / cowl flaps (memory) ..... OFF / CLOSED (overhead vents open)

Airspeed (memory) ..... 100 KIAS (increase if fire persists until incombustible)

Do not attempt an engine restart.

Forced landing ..... EXECUTE (see Engine Failure in Flight)

Electrical Fire in Flight

LAND AS SOON AS POSSIBLE

Standby battery (memory) ..... OFF (G1000)

Master switch (ALT & BAT) (memory) ..... OFF

Vents / cabin air / heat (memory) ..... CLOSED

Fire extinguisher ..... ACTIVATE (if available)

Avionics switch (BUS 1 & BUS 2) ..... OFF

All other switches (except ignition) ..... OFF

After discharging the extinguisher, ventilate the cabin.

▶ **Fire out and electrical power necessary?**

Yes:

Master switch (ALT & BAT) ..... ON

Standby battery ..... ARM (G1000)

Circuit breakers ..... CHECK – do not reset a faulted breaker

Avionics switch (BUS 1 then BUS 2) ..... ON one at a time, with delay, check for smoke

Vents / cabin air / heat ..... OPEN when fire is completely out

Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. ..... NOTED

No:

Continue with master off. No flaps, radios or fuel gauges; standby instruments only (G1000).

Deferred: Electrical power off – flaps-up landing (approach 80 KIAS), no radio; standby instruments only. ..... NOTED

Cabin Fire

LAND AS SOON AS POSSIBLE

Master switch (ALT & BAT) (memory) ..... OFF

Vents / cabin air / heat (memory) ..... CLOSED

Fire extinguisher ..... ACTIVATE

Cabin ..... VENTILATE after fire is out

Land ..... AS SOON AS POSSIBLE, inspect for damage

Emergency Descent

Throttle (memory)	**IDLE**
Propeller (memory)	**HIGH RPM (full forward – maximum drag)**
Mixture (memory)	**RICH**
Wing flaps (memory)	**UP**
Airspeed (memory)	**140 KIAS smooth air / 110 KIAS turbulence (Va)**
Cowl flaps	**CLOSED (avoid shock cooling)**
Power	**CLEAR ENGINE periodically**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 849ad9ab) · 2026-11-07